Raj Kamal Mehta

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Experience

AMSN Syntax IT Solution Pvt. Ltd. | *Mobile & Web Development Intern* Jan 2026 – Apr 2026

- **Tech stacks used:** React Native, React.js, REST APIs
- **Frontend Architecture:** Engineered high-performance, cross-platform architectures using React.js and React Native, translating complex design requirements into seamless user interfaces.
- **UI/UX Optimization:** Debugged and resolved critical layout inconsistencies, significantly improving cross-browser compatibility and elevating overall platform stability.

IIIT Naya Raipur | *Machine Learning (ML) & Deep Learning Trainee* July 2025 – Aug 2025

- **Tech stacks used:** Python, React Native, FastAPI, Kotlin, Scikit-Learn
- **Full-Stack Architecture:** Architected a cardiovascular health application extracting raw PPG optical signals directly from a smartphone camera using custom Kotlin Native hardware bridges.
- **Clinical ML Pipeline:** Built a DSP-filtered Random Forest model using 30+ extracted features, achieving **95% accuracy** deployed via FastAPI.

Projects

MittiCard – Soil Health App | *React Native, PostgreSQL, Groq Vision* | (GitHub) Feb 2026 – Apr 2026

- Developed an AI-powered mobile app that scans ICAR Soil Health Cards using Groq Vision OCR and delivers crop-specific fertilizer recommendations with cost estimates in Hindi and English.
- Engineered an advisory system for 7 crops with NPK deficiency detection and soil health scoring (0–100), supported by a Node.js/Express and PostgreSQL backend with OTP authentication and Firebase push notifications.

Bike Rental App – Booking Platform | *React Native, Node.js, MongoDB, Stripe* | (GitHub) Dec 2025 – Jan 2026

- Architected a cross-platform mobile application supporting end-to-end booking workflows, secure Stripe payments, and automated KYC via REST APIs.
- Designed a scalable Node.js backend with JWT authentication and optimized MongoDB schemas, reducing query response time by **30%** under concurrent load.

PPG Signal Quality Assessment | *React Native, FastAPI, Scikit-Learn, Kotlin* | (GitHub) July 2025 – Aug 2025

- Architected a full-stack cardiovascular health application that extracts raw PPG optical signals directly from a smartphone camera using custom Kotlin Native hardware bridges.
- Engineered a clinical Machine Learning pipeline applying Butterworth DSP filters and extracting 30+ statistical features to achieve **95% classification accuracy** with a Random Forest model deployed on FastAPI.

Education

Krishna Vikas Institute of Technology , Raipur 2022 – 2026

B.Tech, Computer Science and Engineering *CGPA: 8.3/10*

Skills

Languages	JavaScript, TypeScript, Python, C
Frontend	React.js, Redux, HTML5, CSS3, Tailwind CSS
Backend	Node.js, Express.js, REST APIs, JWT Authentication
Mobile Dev	React Native (CLI, Expo)
Databases	MongoDB, PostgreSQL, MySQL
Tools	Git, GitHub, Docker, CI/CD, Postman, Firebase, VS Code

Achievements

Publication: Raj Kamal Mehta et al., "**PPG Signal Quality Assessment & Artifact Detection for Smartphone Monitoring**," *TechRxiv (IEEE-supported preprint)*, 2025.

Certificate: Artificial Intelligence & Machine Learning Training, *Infosys Springboard* (2024)

Certificate: Job-Ready MERN Full-Stack Web Development, *30 Days Coding* (Nov 2024)

Hackathon Finalist: IDEA-FIESTA Hackathon (2022)